

InsightOS - Complete Hackathon Guide (Parts 1-20)

InsightOS - Complete Hackathon Preparation Guide

> **Based on exhaustive analysis of every file in the codebase. No assumptions - every claim is verified from source code.**

PART 1 - COMPLETE PROJECT SUMMARY

1. What is InsightOS?

InsightOS is a **browser-based AI-powered Business Intelligence (BI) platform** designed for multi-store retail operations in India. It ingests CSV data from POS (Point-of-Sale) systems and inventory databases, runs a deterministic rule-based analytics engine to detect business problems (stockouts, low margins, expiry risks), and provides an AI Copilot chat interface powered by **IBM Granite LLM** (via local Ollama or IBM watsonx cloud API) for natural-language business Q&A.

It is a **pure frontend application** (React + TypeScript + Vite) with no dedicated backend server - all data processing, analytics, and AI API calls happen client-side.

2. What real-world problem does it solve?

Small and medium retailers in India rely on Excel spreadsheets, manual stock-checks, and gut instinct to make inventory and pricing decisions. This leads to:

Stockouts - products run out without warning, losing sales

InsightOS converts raw transactional CSV data into prioritized, actionable business alerts with financial impact estimates and step-by-step remediation plans.

3. Who are the target users?

Multi-store retail chain owners (clothing, electronics, grocery)

4. Why is this problem important?

* India has ~14 million retail outlets* (IBEF). The vast majority have zero BI tooling.

*
reduc

5. What is our proposed solution?

A zero-infrastructure, browser-based BI platform that:

1. Acc

6. What are the main features?

| # | Feature | Status |

|---|---

7. What is the complete user workflow?

...

1. Op

8. What happens technically at each major action?

CSV Upload

1. File selected ? PapaParse parses CSV string into array of row objects
2. `csv`

Analytics Computation

1. `analyzeBusinessData()` receives posData + inventoryData + active filters
2. `filter`

AI Chat

1. User types question ? `handleSend()` in AIChat.tsx
2. Build

9. What makes the project different from existing solutions?

Aspect Existing Tools InsightOS	-----
-------------------------------------	-------

10. What is the core innovation?

****Deterministic Business Rules + Generative AI in a zero-backend architecture.****

The analytics engine uses quantitative thresholds (margin <15%, stock ? reorder level, expiry ? 15 days) to generate

reliable, mathematically-grounded alerts. The AI layer (IBM Granite) adds natural-language explanation and strategic reasoning on top. Neither alone is sufficient - rules without AI are rigid and unexplainable; AI without rules can hallucinate numbers.

11. What are the strongest selling points for a hackathon judge?

1. **Fully working end-to-end prototype** - not a mockup, not slides

2. **D**

Pitch Versions

30-Second Pitch

> "Small retailers in India lose lakhs every year to stockouts, expired inventory, and invisible margin problems - but they can't afford Power BI or hire data analysts. InsightOS is a free, browser-based BI platform that takes their existing CSV sales data and instantly shows them what's going wrong, how much money they're losing, and exactly what to do about it - powered by IBM Granite AI for natural-language business intelligence. No servers, no setup, no subscription."

1-Minute Pitch

> "India has 14 million retail outlets, and most of them run on Excel and gut instinct. They have sales data - POS CSVs, inventory spreadsheets - but no way to extract intelligence from it."

>

3-Minute Pitch

> "Let me paint a picture. You own a chain of 5 retail stores across Indian cities. Every month, you get CSV exports from your POS system. You open Excel, stare at thousands of rows, and try to figure out: What sold well? What's running out? Which store is losing money? Are any products about to expire?"

>

5-Minute Pitch

> *[Use the 3-minute version above, then add the following]*

>

PART 2 - COMPLETE TECHNOLOGY STACK

| Technology | Version | Where Used | Why We Used It | Why It Was a Good Choice | Alternatives Considered |

|-----

Technologies NOT Used (Clarification)

| Technology | Status | Explanation |

|-----

PART 3 - IBM TECHNOLOGY EXPLANATION

IBM Granite LLM

What it is

IBM Granite is a family of large language models developed by IBM Research, designed for enterprise AI applications. InsightOS uses two variants:

* **

What role it plays

The AI Copilot tab. When a user types a natural-language business question (e.g., "Which store should I expand?", "Why did revenue drop?"), the question and the full business context dataset are sent to IBM Granite, which generates a

domain-specific, data-grounded response.

Why we selected it

1. **IBM hackathon alignment** - the hackathon specifically encourages IBM technology usage

2. **R**

How it integrates with the rest of the system

...

User C

What would happen if we removed it

The platform would still function for all dashboard analytics, charts, heatmaps, alerts, and data exploration. Only the AI Copilot chat tab would lose functionality - it would display the connection guidance message. **All business intelligence is generated by the deterministic analytics engine, not by AI.** The AI is an augmentation layer, not a dependency.

Why it is better for this project than alternatives

| Factor | IBM Granite | OpenAI GPT-4 | Google Gemini | Meta Llama |

|-----

Limitations

1. **8B local model has limited reasoning depth** compared to 70B+ models

2. **N

How it could scale in production

1. **Deploy Ollama on a GPU server** (NVIDIA A100/H100) for multi-user inference

2. **U

Prompt Engineering Details

The prompt sent to Granite is structured as:

...

This is a form of **in-context learning / RAG** without a vector database - the live analytics are injected directly into the prompt so the model can reason over actual business data.

Hallucination handling

1. **All numerical KPIs on the dashboard come from the deterministic analytics engine, NOT from AI** - so users never see hallucinated numbers in the dashboard

2. **T**
groun

IBM watsonx.ai Cloud Platform

What it is

IBM's enterprise AI platform for deploying and consuming foundation models via REST API. InsightOS uses it as a cloud fallback when local Ollama is unavailable.

Integration details

* **Endpoint**: `https://us-south.ml.cloud.ibm.com/ml/v1/text/generation?version=2023-05-29`

* **

PART 4 - SYSTEM ARCHITECTURE

Architecture Overview

InsightOS is a **pure client-side Single Page Application (SPA)** with no dedicated backend. The architecture is:

...

?????

PowerPoint-Ready Architecture Diagram

****Simple version (left-to-right):****

...

Key Architectural Decisions

1. ****No backend**** - Eliminates server costs, simplifies deployment, keeps data private on user's machine

2. ****V**

PART 5 - DATA ? INSIGHT PIPELINE

> ****Note:** InsightOS does NOT support image upload or OCR. Data input is exclusively via CSV files. The pipeline below describes the actual CSV ? Insight flow.

CSV ? Insight Pipeline

Stage 1: File Selection

* User clicks "Upload CSV" button ? modal opens
 ([CSVUpload.tsx](file:///c:/Users/abhay/OneDrive/Desktop/hackathon/src/components/CSVUpload.tsx))

* T

Stage 2: CSV Parsing (PapaParse)

* `Papa.parse(file, { header: true, skipEmptyLines: true })` converts CSV text into array of objects

* E

Stage 3: Header Normalization ([csvNormalizer.ts](file:///c:/Users/abhay/OneDrive/Desktop/hackathon/src/uti

* `cleanKey(key)` ? strips all non-alphanumeric characters, lowercases: `"Sales Amount (Rs. )"` ? `"salesamount"``

* M

Stage 4: Data Cleaning & Defaults

* Missing `date` ? current date in `YYYY-MM-DD` format

* M

Stage 5: Analytics Engine ([dataAnalysis.ts](file:///c:/Users/abhay/OneDrive/Desktop/hackathon/src/utils/dataAnalysis.ts))

* `analyzeBusinessData()` processes all data through:

1. D

Stage 6: Business Intelligence Output

* Returns `BusinessInsight` object containing:

- 5 K

Data Quality Protections

| Issue | How It's Handled |

|-----

PART 6 - BUSINESS LOGIC

Inventory Intelligence

Low-Stock Detection ([x] Implemented)

* **Rule**: `currentStock <= reorderLevel` and `currentStock > 0`

* **

Stockout Detection ([x] Implemented)

* **Rule**: `currentStock === 0`

* **

Expiry Risk Detection ([x] Implemented)

* **Critical (?15 days)**: 40% flash clearance discount. Capital recovery = `stock × unitCost × 0.6`

* **

Overstock Detection ([] Not Implemented - Future Feature)

Inventory Trends Over Time ([] Not Implemented - Future Feature)

Supplier Performance Scoring ([] Not Implemented - Future Feature)

Stock Movement Velocity ([x] Partially Implemented)

* Days of Supply = `currentStock / (totalQuantity / 30)` * calculated per product in ABC matrix

Sales / POS Intelligence

Revenue Trends ([x] Implemented)

* Monthly aggregation in `calculateMonthlySales()`, rendered as AreaChart (revenue + profit curves)

* T

Best-Selling Products ([x] Implemented)

* ABC Classification: Class A = cumulative top 80% of revenue (hero products)

* D

Poor-Performing Products ([x] Implemented)

* Class C = bottom 5% of cumulative revenue

Category Performance ([x] Implemented)

* Revenue breakdown by category with share-of-revenue percentage

Time-Based Trends ([x] Partially Implemented)

* Monthly aggregation exists. Daily granularity if data spans single month.

Customer Buying Patterns ([] Not Implemented - Future Feature)

Business Recommendations

Products to Restock ([x] Implemented)

* Stockout and low-stock alerts with supplier name and reorder quantity

Products to Promote ([x] Implemented)

* Hero Product opportunity identifies Class-A top revenue generator

Products Underperforming ([x] Implemented)

* Class-C product identification

Products Sold at a Loss ([x] Partially Implemented)

* Low margin detection: `profitMargin < 15%` at store level

Poor-Performing Suppliers ([] Not Implemented - Future Feature)

Potential New Products ([] Not Implemented - Future Feature)

Staffing Requirements ([] Not Implemented - Future Feature)

Expansion Opportunities ([x] Implemented)

* High-growth stores (>15% growth rate) flagged as expansion opportunities

InsightOS - Parts 9-15 (Competitor, Security, Scale, Limitations, Roadmap,

PART 9 - COMPETITOR ANALYSIS

Feature Comparison Matrix

| Feature | InsightOS | Excel/Sheets | Power BI | Tableau | POS Systems (Petpooja, Marg) | ChatGPT/AI Assistants |

Why a Business Would Choose InsightOS

- 1. **vs Excel:** InsightOS automates what would take hours of pivot tables, VLOOKUP formulas, and conditional formatting. A retailer doesn't need to know how to build a pivot table to see that their Green Tea is expiring in 9 days.
- 2. **vs Power BI:** Power BI requires an Azure subscription, a database connection, and someone who knows DAX formulas to build reports. InsightOS is pre-built - upload CSV, see intelligence. Different market segments: Power BI for enterprises with BI teams, InsightOS for SMBs with no technical staff.
- 3. **vs Tableau:** Same issue as Power BI - powerful but requires expertise. Tableau is a visualization platform; InsightOS is an intelligence platform. We don't just show data - we detect problems and recommend actions.
- 4. **vs POS Systems:** POS systems generate the data but don't analyze it. InsightOS sits downstream of any POS system, consuming its CSV exports and adding the intelligence layer that POS vendors don't provide.
- 5. **vs ChatGPT:** ChatGPT doesn't have your data, doesn't persist across sessions, can't show interactive charts, can't run deterministic business rules, and can't display a geospatial heatmap. InsightOS is a purpose-built platform, not a chat window. Also, ChatGPT requires sharing sensitive business data with OpenAI servers.

PART 10 - SECURITY & PRIVACY

Current Security Posture

Authentication

Status: [] Not implemented

Risk

Authorization

Status: [] Not implemented (no user roles)

**Risk

Data Privacy

Status: [x] Strong by architecture

**Rea
using

API Key Exposure

Status: [!] Known vulnerability

**Risk
DevTo

Input Validation (CSV)

Status: [x] Implemented

**Deta
missin

XSS (Cross-Site Scripting)

Status: [x] Protected by React

**Deta
tags. V

CSRF (Cross-Site Request Forgery)

Status: N/A

**Deta

SQL Injection

Status: N/A

**Deta

AI Data Privacy

Status: [!] Depends on inference path

Summary of Security Weaknesses

| Weakness | Severity | Production Fix |

|-----

PART 11 - SCALABILITY

Current Prototype Capacity

| Dimension | Current Capacity | Bottleneck |

|-----

Scaling Plan

10 Businesses (Hackathon ? MVP)

Frontend: Deploy static site on Vercel (free tier)

100 Businesses (MVP ? Early Growth)

Frontend: Vercel Pro with CDN

10,000 Businesses (Growth ? Scale)

* **Frontend:* * Multi-region CDN with edge caching

* **

1,000,000 Businesses (Enterprise Scale)

* **Frontend:* * Global CDN (CloudFront) with ISR (Incremental Static Regeneration)

* **

PART 12 - LIMITATIONS (Honest Assessment)

Current Limitations

| # | Limitation | Severity | Impact | Realistic Fix |

|---|---

How to Answer "What are your limitations?" to a Judge

> "We're transparent about our prototype's boundaries. The three biggest limitations are: (1) no data persistence - if you refresh the browser, data resets to sample data, (2) no authentication - this is a single-user prototype, and (3) AI response latency on CPU can be 15-45 seconds.

>

PART 13 - FUTURE ROADMAP

Phase 1 - Hackathon Prototype [x] (Current State)

| Feature | Status |

|-----

Phase 2 - MVP (2-4 weeks after hackathon)

| Feature | Priority | Effort |

|-----

Phase 3 - Production (1-3 months)

| Feature | Priority |

|-----

Phase 4 - Scale (3-12 months)

| Feature | Priority |

|-----

PART 14 - DEMO SCRIPT (3-5 Minutes)

Setup Before Demo

* Browser open to `http://localhost:5173/`

* O

Step 1: Problem Statement (30 seconds)

What I say:

> "Ima
with th
runnin
you ca

What's on screen: InsightOS dashboard with KPI cards visible.

Tech

Step 2: Solution Introduction (15 seconds)

What I say:

> "Ins
losing

What I do: Point to the screen showing the dashboard.

Tech

Step 3: Dashboard Overview (45 seconds)

What I do: Point to each KPI card.

What I say:

> "He

Technical concept: Deterministic KPI computation, risk status classification.

Step 4: Alert Center & Remediation (60 seconds)

What I do: Scroll down to alerts. Click on an expiry alert.

What I say:

> "The
risk al

****What I do:**** Close the modal.

****Tech**
steps.

Step 5: Charts & Heatmap (45 seconds)

****What I do:**** Click "Analytics Studio" tab. Show the revenue trend chart. Then click "Store Heatmap" tab.

****What I say:****

> "Ou
break

****Technical concept:**** Recharts data visualization, Leaflet geospatial mapping, CircleMarker radius scaling.

Step 6: AI Copilot (60 seconds)

****What I do:**** Click "AI Copilot" tab. Type a question.

****What I say:****

> "No
Ollam

****What I type:**** "Which store should I invest in for expansion and why?"

****What I say (while waiting):****

> "No
margi
our ac

****What I do:**** When response appears, read the key recommendation.

> "It identified our highest-growth store and explained why, using actual numbers from our dataset. This is data-grounded AI, not a generic chatbot."

****Technical concept:**** IBM Granite LLM, contextual prompt engineering, in-context RAG, local inference via Ollama.

Step 7: Data Upload (30 seconds)

What I do: Click "Upload CSV" button. Show the modal.
What I say:

Technical concept: PapaParse CSV parsing, fuzzy header normalization with alias arrays.

Step 8: Closing (15 seconds)

What I say:

PART 15 - PRESENTATION STRUCTURE (10 Slides)

Slide 1: Title (15 seconds)

Title: * InsightOS * AI-Powered Business Intelligence for Indian Retail

Slide 2: The Problem (45 seconds)

Title: * The Rs. 92,000 Crore Problem

Slide 3: The Solution (45 seconds)

Title: * InsightOS: Upload CSV ? See Intelligence

Slide 4: How It Works - Architecture (30 seconds)

* **Title:* * Zero-Backend Architecture

* **

Slide 5: Key Features - Intelligence Engine (60 seconds)

* **Title:* * Automated Business Intelligence

* **

Slide 6: Technology Stack & IBM Integration (45 seconds)

* **Title:* * Built on IBM Granite + Modern Web Stack

* **

Slide 7: Live Demo (120 seconds)

* **Title:* * Live Demo

* **

Slide 8: Competitive Advantage (30 seconds)

* **Title:* * Why InsightOS Wins

* **

Slide 9: Roadmap & Business Impact (30 seconds)

* **Title:* * From Prototype to Platform

* **

Slide 10: Closing (15 seconds)

* **Title:* * InsightOS * Intelligence, Not Just Data

* **

PART 16 - 30-SECOND ELEVATOR PITCH

"Hi! We built **InsightOS** - a zero-backend, AI-powered Business Intelligence platform designed specifically for Indian multi-store retail owners.

Small and medium retailers in India lose lakhs every year to out-of-stock items, expired inventory, and low margins. They have raw CSV sales data, but they can't afford expensive tools like Power BI or hire data analysts.

InsightOS solves this. You open your browser, upload your POS and inventory CSVs, and within seconds, our engine detects every operational warning, computes the exact financial loss in rupees, and gives you step-by-step action plans to recover capital.

On top of that, we integrated an AI Copilot powered by **IBM Granite** that lets you ask questions like 'Which store should I expand?' or 'Why are my margins down?' and get data-grounded answers immediately. It is free, private, and runs entirely in the browser."

PART 17 - TECHNICAL DEEP DIVE

1. Fuzzy CSV Normalizer (`csvNormalizer.ts`)

Problem: Every POS system (Tally, Petpooja, Marg ERP) exports CSVs with different header names. A general user shouldn't have to rename columns manually.

2. Browser-Only Analytics Engine (`dataAnalysis.ts`)

Problem: Running calculations on thousands of transaction rows in the frontend can freeze the UI.

3. Dual-Path AI Inference & Token Caching (`aiService.ts`)

Problem: Cloud APIs cost money and risk data leaks. Local inference requires setup and is slower.

PART 18 - "WHY DID YOU CHOOSE X?" CHEAT SHEET

Why React 19? React's component model maps naturally to dashboard tabs. React 19's automatic batching and fast reconciliation make UI rendering efficient when filters change.

PART 19 - PERSONAL CONTRIBUTION QUESTIONS

* **What did you personally build?* * I built the fuzzy CSV normalizer, the core deterministic analytics engine (`dataAnalysis.ts`), the Smart Expiry Discounting logic, and the dual-path IBM Granite AI integration (Ollama + watsonx cloud fallback).

*
Ollam

PART 20 - FINAL CHEAT SHEET (HACKATHON SURVIVAL SHEET)

Platform DNA

* **One-sentence description:* * InsightOS is a zero-backend, AI-powered Business Intelligence platform that converts raw POS and inventory CSVs into prioritized operational alerts and natural-language insights via IBM Granite.

*
client-

Core Metrics & Formulas

* **Days to Expiry:* * `(expiryDate - referenceDate) / (1000 \* 60 \* 60 \* 24)`

* **

Ranked Judge Questions

1. [!] Why use InsightOS instead of Excel?

Answer: Excel shows data, but you have to build the logic yourself. InsightOS automatically normalizes your CSV, runs 12 retail business rules, identifies stockouts, estimates rupee losses, and suggests clearance discounts based on expiry dates. Plus, it has an AI Copilot that answers questions in plain English.

2. [!] Isn't this just an AI wrapper?

Answer: No. If you disconnect the AI, the dashboard, alerts, KPI cards, interactive charts, and Leaflet maps still function. 80% of the value is generated by our deterministic rules engine. The AI is an overlay for natural language Q&A.

3. [!] What happens if your AI hallucinates?

Answer: The dashboard KPIs, margin calculations, and inventory warnings are calculated mathematically by our rules engine - they can never hallucinate. The AI only operates in the chat interface, and we ground it by injecting computed business metrics directly into the prompt.

4. [!] Why did you choose IBM Granite?

Answer: Granite is built for enterprise factual accuracy, making it perfect for business intelligence. It has a small footprint (`granite3-dense:8b` is 4.9GB), allowing us to run it locally on consumer hardware for free. If local inference fails, it seamlessly falls back to watsonx cloud API.

5. How does the data stay private?

Answer: InsightOS has no backend database. All parsing, cleaning, and analytics happen in browser memory. When running local Ollama, even the AI model runs offline. Your data never leaves your computer.

6. How would you scale to 1,000 stores?

Answer: While the analytics engine processes 100K rows in the browser in under 500ms, scaling to large-scale enterprises would require adding a backend API server (Node.js/FastAPI), a persistent database (PostgreSQL), and webhooks to integrate directly with POS APIs.

7. How does the CSV normalizer handle custom formats?

Answer: It uses fuzzy matching. It cleans all headers (strips non-alphanumeric, lowercases) and checks against alias arrays. "Sales Amount", "MRP", and "Revenue" all map to the canonical "revenue" field.

8. What is the business model?

Answer: A freemium SaaS model. Single-store basic analytics is free. Multi-store dashboards, advanced PDF reporting, and the AI Copilot are part of a paid tier at Rs. 999-2,999/month.

9. How do you compute financial loss for stockouts?

Answer: We estimate lost revenue per period as the product's average selling price multiplied by its reorder level. If no pricing history exists, we use unit cost times 10 as a conservative placeholder.

10. [!] What parts did you build vs AI-generated?

Answer: I wrote the core business logic, the CSV normalizer, and the prompt injection templates. AI helped accelerate boilerplate creation for charts and maps. I understand and can explain every line of code in the repository.

10 Technical & AI Cheat Questions

1. **How do you prevent CORS issues with local Ollama?**

- Co